

Comptia A+ Assignment

[A+ - Peripherals and Power Supply]

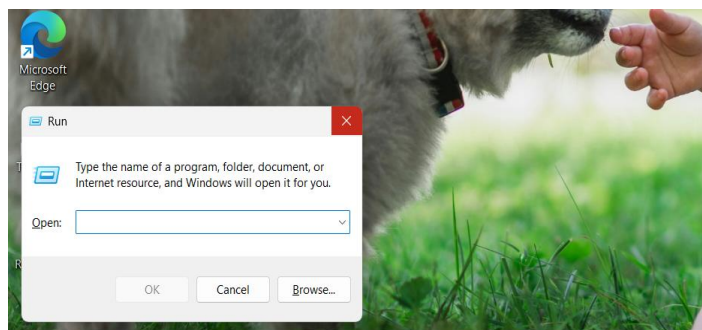
Operating System information, windows, Linux, macOS installation and adjust OS settings.

Task

❖ **1. On your windows laptop or PC, open 'System Information' and take a screenshot showing your OS version, processor, and RAM details.**

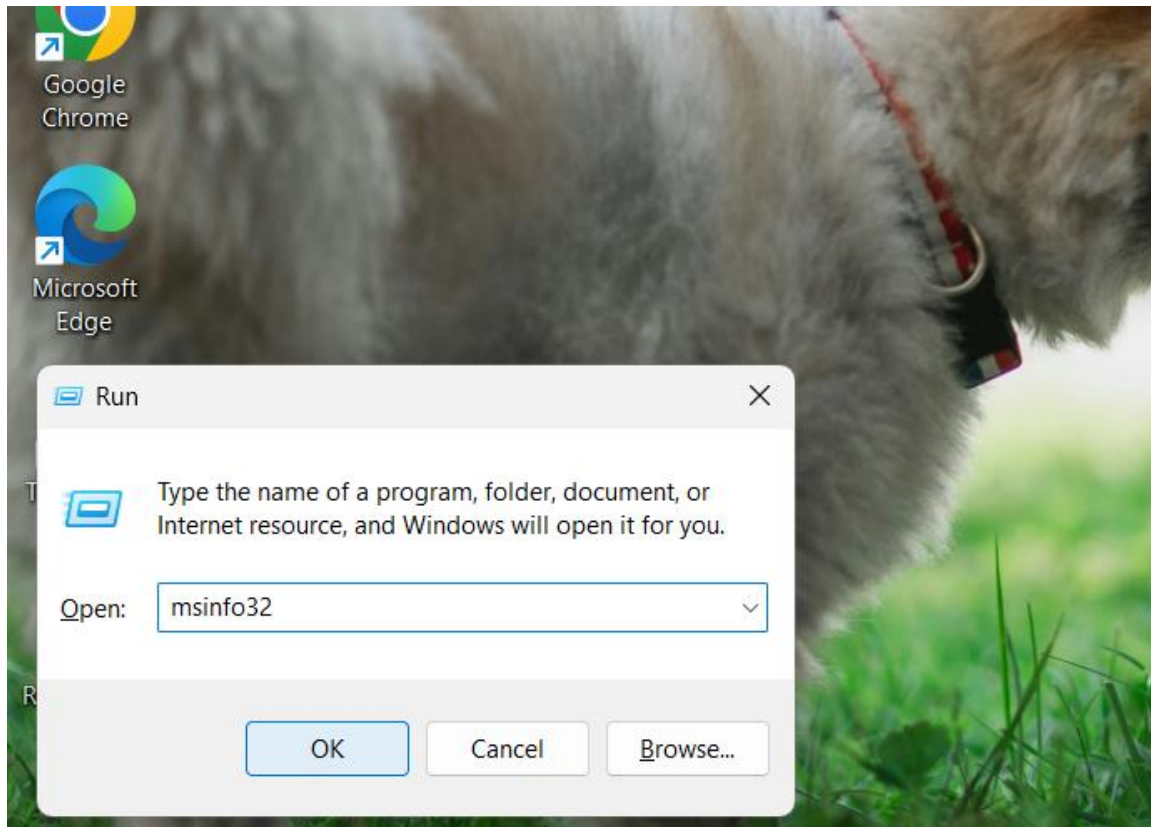
➤ Operating System Information and Viewing System Information in Windows

➤ Step 1: Open the Run Dialog:



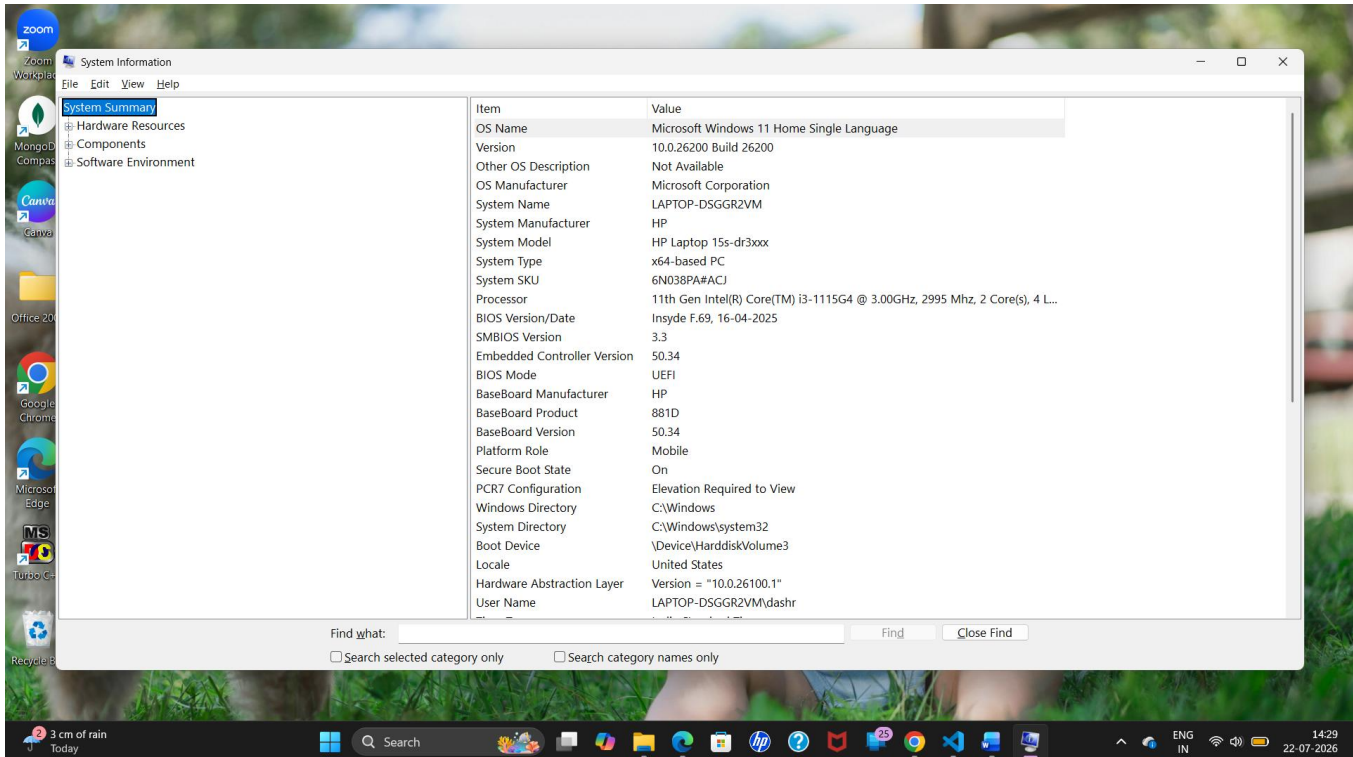
Press Windows + R to open the Run dialog box. This shortcut provides quick access to Windows utilities and administrative tools.

➤ Step 2: Type msinfo32:



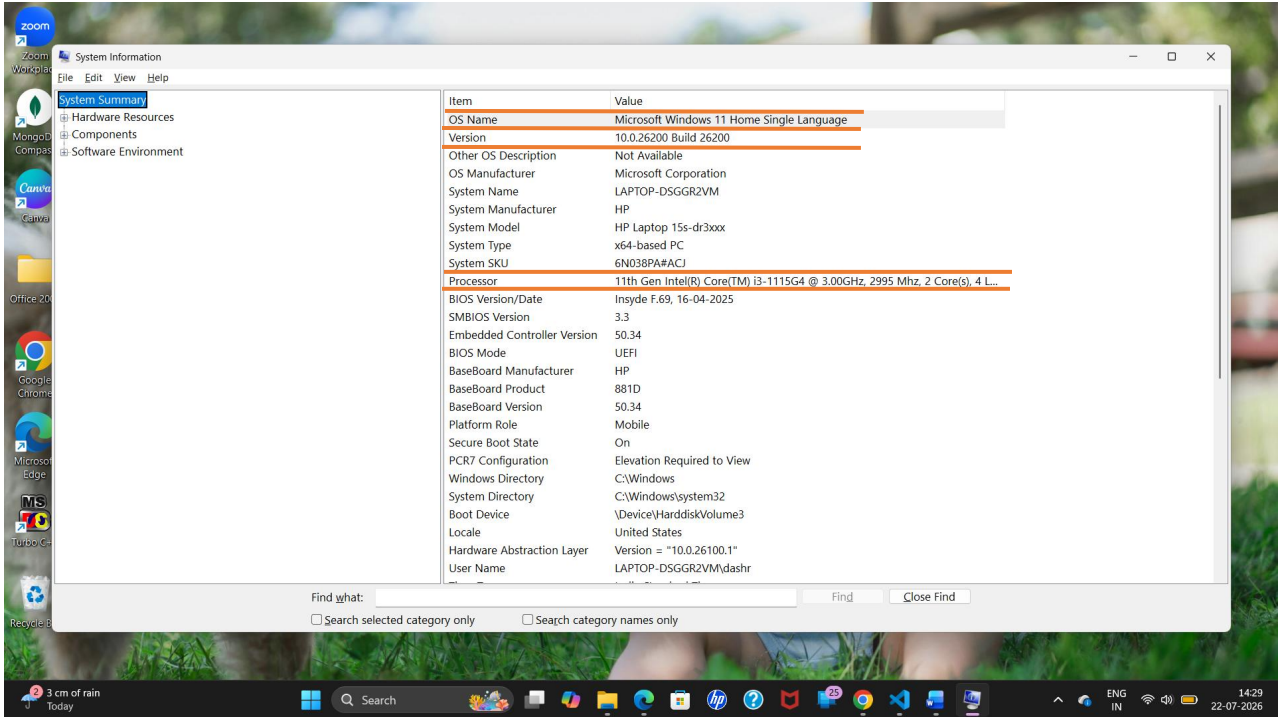
Type msinfo32 in the Run dialog box and click OK. This command opens the System Information utility.

➤ Step 3: Open the System Information Window:



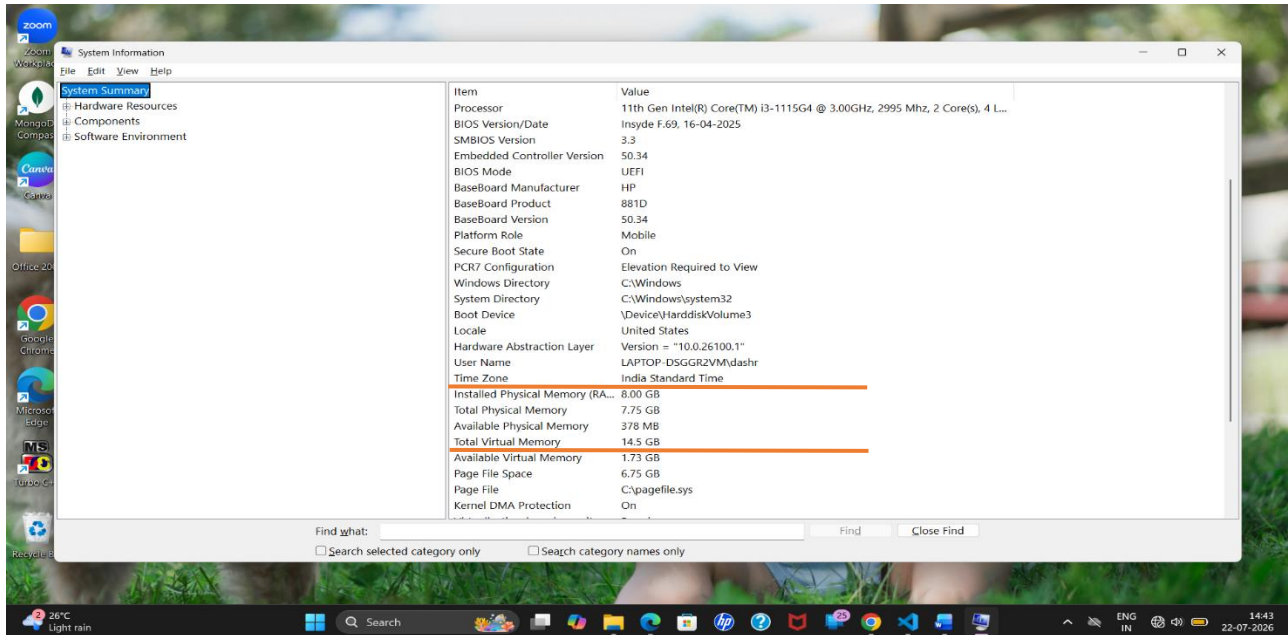
The System Information window displays important system details such as the operating system version, processor, installed RAM, BIOS version, and system type.

➤ Step 4: Verify OS Version, Processor and RAM:



The screenshot shows the Windows System Information window. The left sidebar has 'System Summary' selected. The main pane displays a table of system information. The 'OS Name' is 'Microsoft Windows 11 Home Single Language' and the 'Version' is '10.0.26200 Build 26200'. The 'Processor' is '11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz, 2995 Mhz, 2 Core(s), 4 L...'. The taskbar at the bottom shows the date and time as 14:29 on 22-07-2026.

Item	Value
OS Name	Microsoft Windows 11 Home Single Language
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	LAPTOP-DSGGR2VM
System Manufacturer	HP
System Model	HP Laptop 15s-dr3xxx
System Type	x64-based PC
System SKU	6N038PA#ACJ
Processor	11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz, 2995 Mhz, 2 Core(s), 4 L...
BIOS Version/Date	Insyde F.69, 16-04-2025
SMBIOS Version	3.3
Embedded Controller Version	50.34
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	881D
BaseBoard Version	50.34
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\Windows
System Directory	C:\Windows\system32
Boot Device	\Device\HarddiskVolume3
Locale	United States
Hardware Abstraction Layer	Version = "10.0.26100.1"
User Name	LAPTOP-DSGGR2VM\dashr



The screenshot shows the Windows System Information window, scrolled down to the 'Memory' section. The 'Installed Physical Memory (RAM)' is 8.00 GB, 'Total Physical Memory' is 7.75 GB, and 'Available Physical Memory' is 378 MB. The 'Total Virtual Memory' is 14.5 GB and 'Available Virtual Memory' is 1.73 GB. The taskbar at the bottom shows the date and time as 14:43 on 22-07-2026.

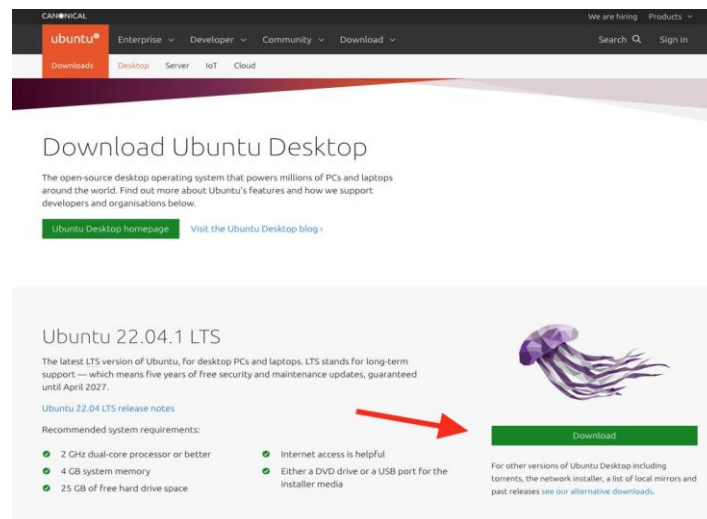
Item	Value
Processor	11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz, 2995 Mhz, 2 Core(s), 4 L...
BIOS Version/Date	Insyde F.69, 16-04-2025
SMBIOS Version	3.3
Embedded Controller Version	50.34
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	881D
BaseBoard Version	50.34
Platform Role	Mobile
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\Windows
System Directory	C:\Windows\system32
Boot Device	\Device\HarddiskVolume3
Locale	United States
Hardware Abstraction Layer	Version = "10.0.26100.1"
User Name	LAPTOP-DSGGR2VM\dashr
Time Zone	India Standard Time
Installed Physical Memory (RAM)	8.00 GB
Total Physical Memory	7.75 GB
Available Physical Memory	378 MB
Total Virtual Memory	14.5 GB
Available Virtual Memory	1.73 GB
Page File Space	6.75 GB
Page File	C:\pagefile.sys
Kernel DMA Protection	On

Verify the Operating System Version, Processor, and Installed Physical Memory (RAM) in the System Summary section. These details confirm the computer's hardware and operating system configuration.

❖ **2. Download any official Linux distribution ISO (like Ubuntu or Fedora), and use Rufus or BalenaEtcher to create a bootable USB drive for installation. If you don't have a spare USB, use a virtual machine tool like VirtualBox to mount the ISO and start the installation process virtually.**

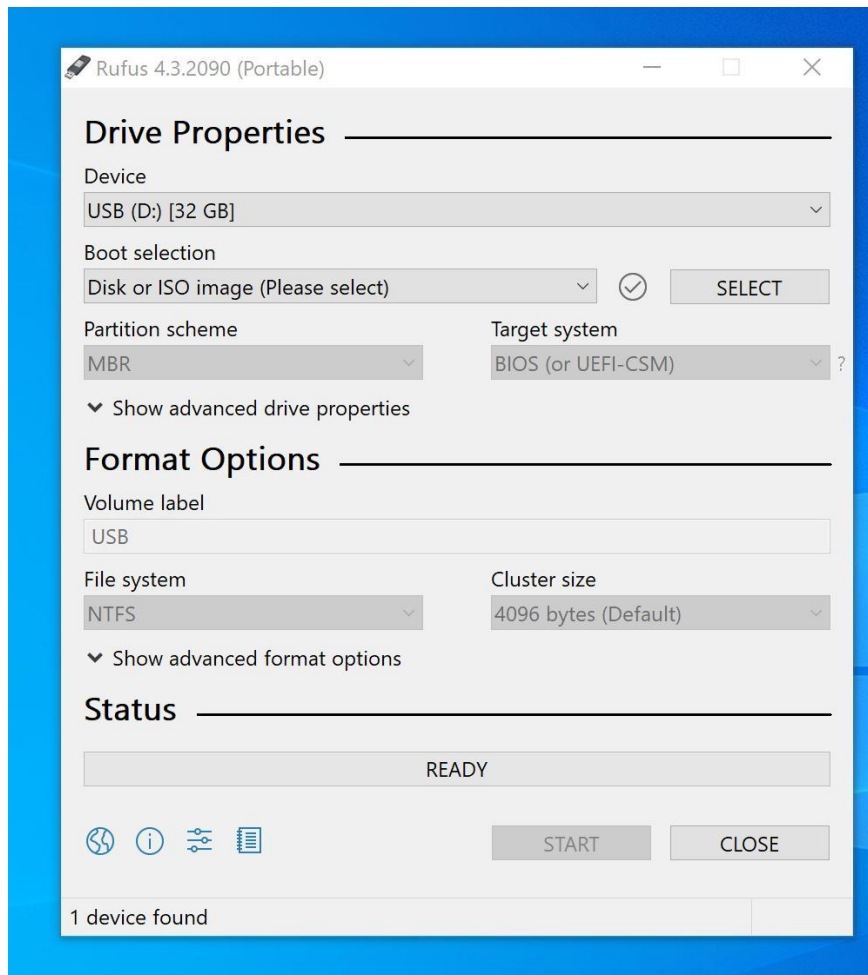
➤ Operating System Installation and Creating a Bootable USB Drive Using Ubuntu ISO and Rufus:

➤ Step 1: Download the Ubuntu ISO:



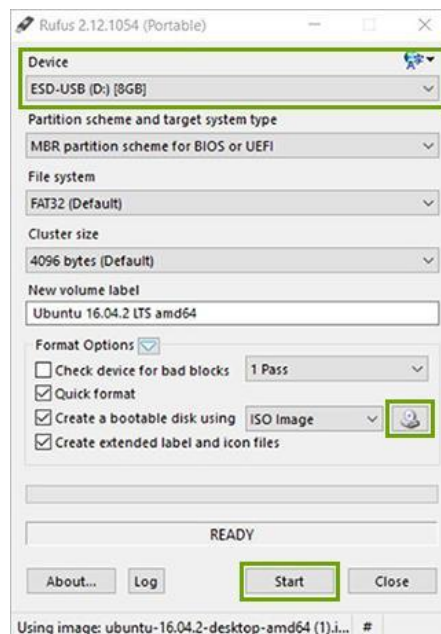
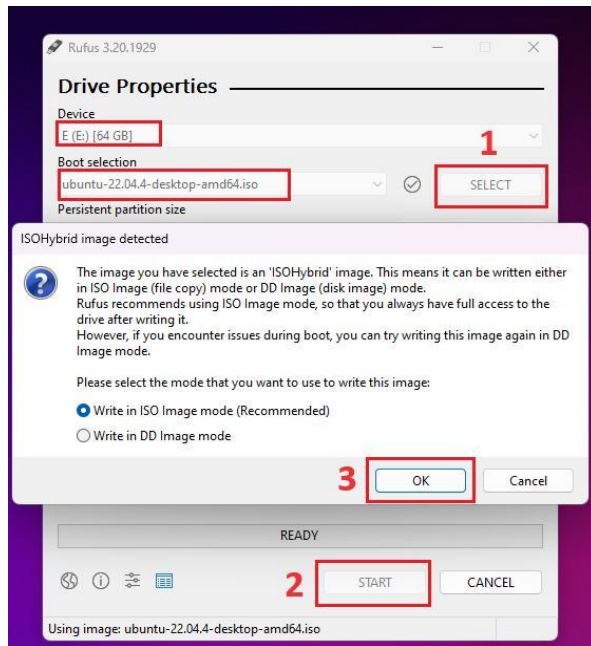
I visited the official Ubuntu website and downloaded the latest Ubuntu Desktop ISO image file for installation.

➤ Step 2: Download and Open Rufus:



I downloaded and opened Rufus, a free utility used to create bootable USB drives from ISO image files.

➤ Step 3: Insert the USB Drive and Select the ISO:



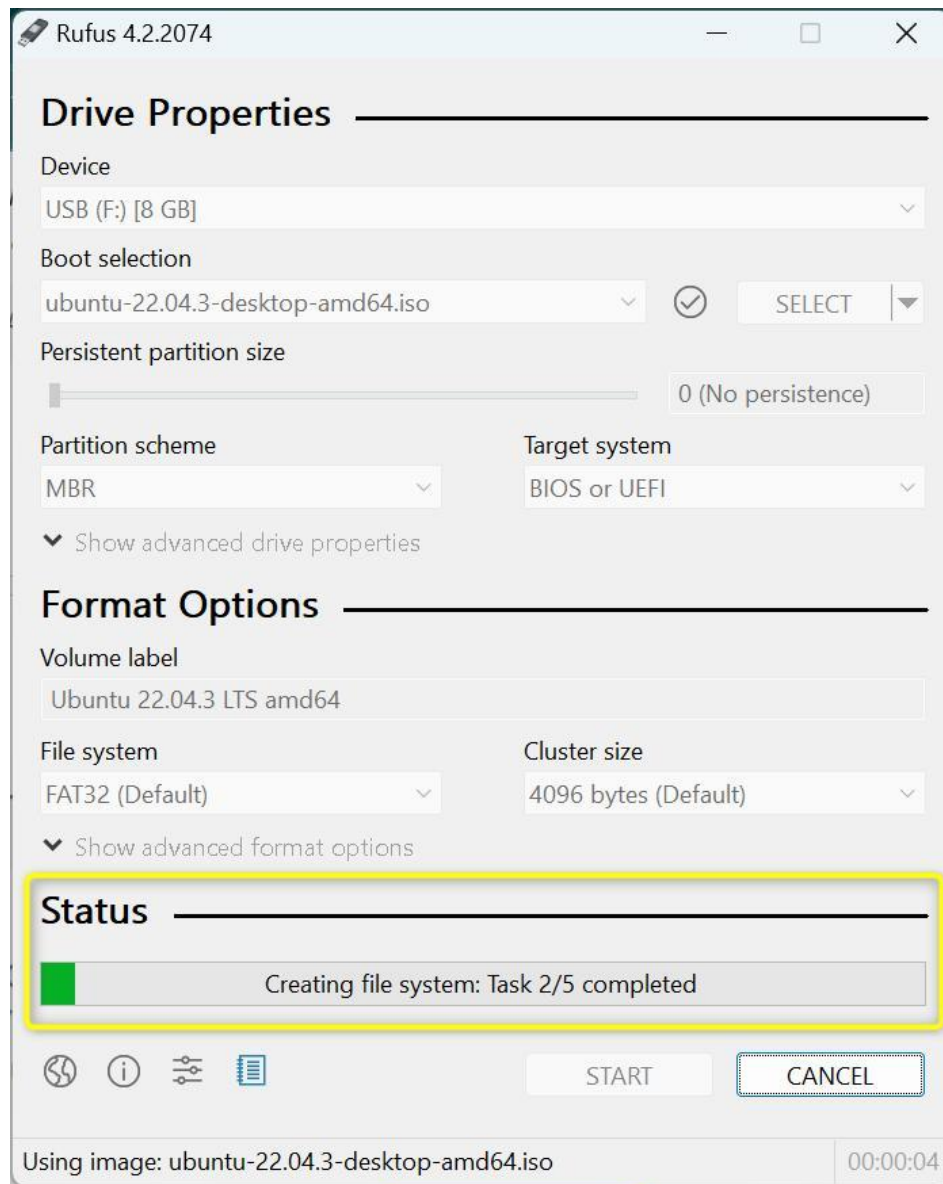
Choose USB key

Choose ISO file

Start the process

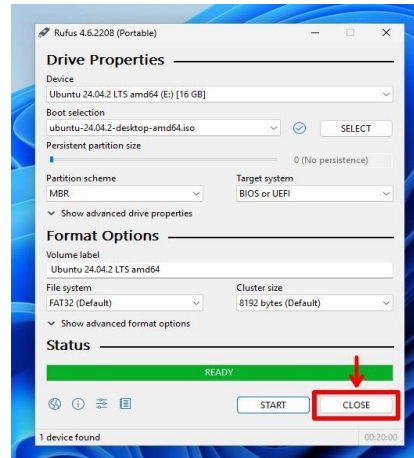
I inserted an 8 GB or larger USB drive, selected it in Rufus, and chose the downloaded Ubuntu ISO using the SELECT button.

➤ Step 4: Create the Bootable USB:



I clicked START in Rufus to create the bootable USB drive. Rufus formatted the USB drive and copied the Ubuntu installation files successfully.

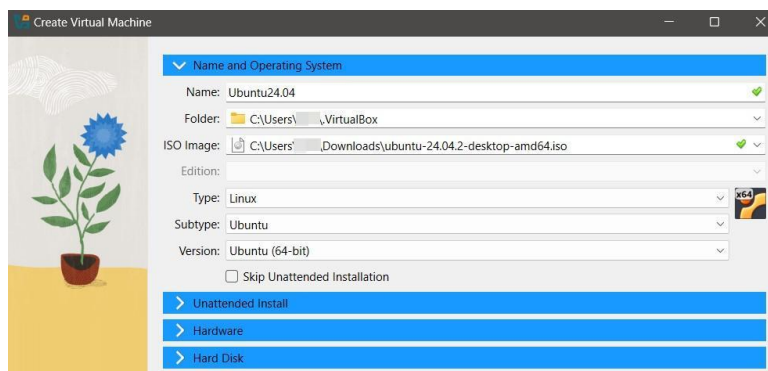
➤ Step 5: Verify the Bootable USB:



After the process completed successfully, the USB drive was ready to boot and install Ubuntu on a compatible computer.

➤ Using VirtualBox:

➤ Use this if you do not have a spare USB drive.

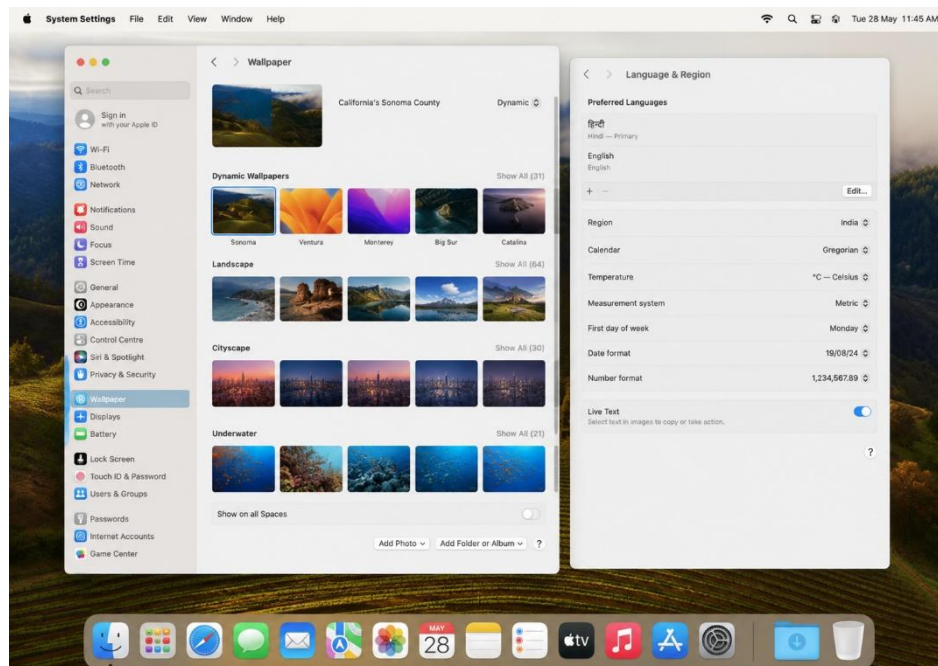


I created a new virtual machine in VirtualBox, mounted the Ubuntu ISO file as a virtual optical disk, and started the installation process without using a physical USB drive.

❖ **3. On a macOS device (or using a macOS virtual machine), open 'System Preferences' and change the desktop wallpaper, then adjust the system language to Hindi or Gujarati. Find 'Language & Region' in System Preferences to add or change languages.**

➤ Changing Wallpaper and System Language in macOS:

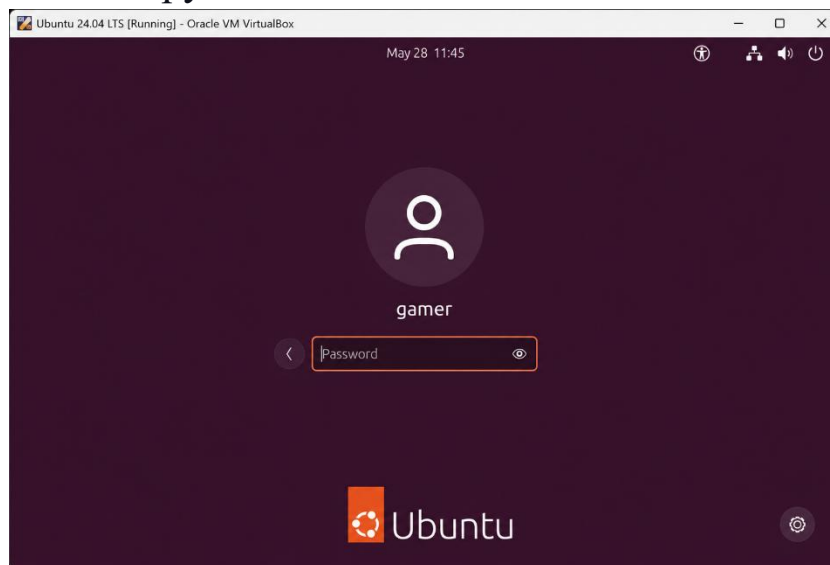
➤ Open System Settings and Change Wallpaper & Language:



System Settings in macOS showing the Wallpaper and Language & Region options. The desktop wallpaper was changed successfully, and the system language was updated to Hindi (or Gujarati) using the Language & Region settings.

- ❖ 4. Install a fresh copy of Ubuntu Linux (real or virtual machine) and create a new user account named 'gamer'. After installation, enable automatic login for this user and take a screenshot of the login screen showing the username.

➤ Install a fresh copy of Ubuntu Linux:



➤ **Procedure**

- Install Oracle VM VirtualBox on Windows.
- Download the official Ubuntu ISO file.
- Create a new virtual machine in VirtualBox.
- Mount the Ubuntu ISO file.
- Start the virtual machine and complete the Ubuntu installation.
- Create a user account named gamer.
- Enable Automatic Login from Settings → Users.
- Restart the virtual machine and verify that the gamer user appears on the login screen.

❖ **5. Use ChatGPT to generate a step-by-step guide for installing Windows 10 on a laptop, including partitioning tips and post-installation settings for privacy and updates. Follow the guide and note any steps that were unclear or missing.**

➤ Using ChatGPT to Generate a Windows 10 Installation Guide:

➤ To generate a step-by-step guide for installing Windows 10 using ChatGPT, including partitioning recommendations, privacy settings, and Windows Update configuration. The guide was reviewed to identify any unclear or missing steps.

➤ Step 1: Download Windows 10

- Visit the official Microsoft website.
- Download the Windows 10 Media Creation Tool.
- Create a bootable USB drive (minimum 8 GB).

➤ Step 2: Boot from the USB Drive

- Insert the bootable USB into the laptop.
- Restart the computer.
- Press the Boot Menu key (F12, Esc, F9, or Del, depending on the manufacturer).
- Select the USB drive as the boot device.

➤ Step 3: Start Windows Installation

- Choose the preferred Language, Time, and Keyboard Layout.
- Click Install Now.

- Enter the product key or choose I don't have a product key if installing later.
- Step 4: Select Installation Type
 - Choose Custom: Install Windows only (Advanced).
- Step 5: Partition the Hard Drive
 - Delete old partitions only if all important data has been backed up.
 - Create a new partition for Windows (recommended 100–150 GB or more).
 - Leave remaining space for personal files.
 - Select the Windows partition and click Next.
- Partitioning Tip:
 - Keep the operating system on a separate partition from personal files.
 - Back up important data before deleting or formatting partitions.
- Step 6: Complete Installation
 - Windows copies files and restarts several times.
 - Do not remove the USB drive until installation is complete.
- Step 7: Configure Initial Settings
 - Select your country and keyboard layout.
 - Connect to a Wi-Fi network.
 - Sign in with a Microsoft account or create a local account.

➤ Step 8: Configure Privacy Settings

- Turn Location off if not required.
- Disable unnecessary Diagnostic Data.
- Turn off Advertising ID if preferred.
- Review all privacy options before continuing.

➤ Step 9: Install Windows Updates

- Open Settings → Update & Security → Windows Update.
- Click **Check for Updates**.
- Install all available updates and restart the computer if required.

➤ Guide Review

- The generated guide was clear and easy to follow. However, a few additional points could improve it:
- It should remind users to back up important files before installing Windows.
 - It should mention the minimum system requirements for Windows 10.
 - It could explain how to enter BIOS/UEFI if the USB does not boot automatically.
 - It should recommend installing device drivers after Windows installation.
 - It could suggest creating a System Restore Point after setup is complete.